

NIGEL VAN DER HORST

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I am a driven NUS Electrical Engineering undergraduate with a passion for embedded systems, computer networking, and cybersecurity. I have hands-on experience across diverse internships - automating data integration into SCADA at Shopee, building an MVP risk-analysis platform at Cysecure, and deploying call-centre networks for Terra Systems – which have sharpened my problem-solving and operational skills. Proactive and collaborative, I excel at turning technical challenges into practical solutions.

EDUCATION

National University of Singapore Bachelor of Engineering (Honours) - Electrical Engineering with a Minor in Computing	Aug 2022 – Dec 2025
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- GPA: 4.03/5.0
- Relevant coursework: Introduction to Programming methodology (python), Microcontroller Programming and Interfacing, Computer networks

Singapore Polytechnic Diploma in Computer Engineering	Apr 2017 - May 2020
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- GPA: 3.64/4.0 (Director's Honour Roll 2019/2020)
- Relevant coursework: Computing Networking, Server Management, Network Security, Cyber Security

WORK EXPERIENCE

Shopee Operation Automation Intern	Jan 2025 - Apr 2025
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- Designed and deployed Netdata for network monitoring across 4 sorting centres, **reducing investigation downtime by 30%**
- Leveraged AWS Lambda and DynamoDB to automate 7 critical alert integrations into native Seataalk app, **reducing manual intervention and incident response time by 20%**
- Integrated network metrics into in-house SCADA systems, enabling correlation of OT/IT issues

Cysecure Pte Ltd Systems Engineer Intern	Jun 2024 - Aug 2024
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- Spearheaded development of an MVP risk analysis platform for non-technical personnel, enabling cost analysis for risks identified

Terra Systems Network Engineering Intern	May 2019 - Oct 2019
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- Troubleshoot network issues for 10+ clients, **maintaining 99.99% SLA compliance**
- Set up 2 call centres using Cisco UCM/UCCX and hardened security via firewall policies

LEADERSHIP ROLES

NUS College of Design and Engineering (CDE) Club, Content Director	Sep 2022 - Sep 2023
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- Oversaw building and bolstering CDE brand identity as a new faculty of NUS for over 2,000 students
- Led a team of 5 to create publicity, growing social media platforms leading to a 65% boost in followers

ACADEMIC PROJECTS

Embedded Systems Developer, STM32 Microcontroller Project	Sep 2023 - Nov 2023
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- Developed a multi-mode embedded system in C on STM32L475 using HAL libraries, interfacing 6 sensors via I2C, UART and GPIO
- Implemented IRQ interrupt handling, real-time telemetry using LCD feedback with custom bitmap graphics

ADDITIONAL INFORMATION

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- **Languages:** English – Native proficiency, Chinese - intermediate working proficiency
 - **Programming:** Python, Assembly, C
 - **Cisco:** Networking, Cyber Security, Collaboration
 - **System admin:** MySQL, Microsoft Windows Server Management, Linux Servers
 - **Cloud/DevOps:** AWS Lambda, DynamoDB, Linux, Netdata